FNLP
Classification – Log-linear Models

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Outline

1. **Feature Representations**
   - Features in NLP
   - Bag-of-Words
2. **Feature based Discriminative Models**
   - Log-Linear Models
   - Model Training
   - Parameter Estimations
   - Regularization

Recall in WSD

We have considered the following clues as useful:

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- syntax clues
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(1) Design those hints
(2) Weight those hints
(3) Do the scoring
(4) Rank the candidates!

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New View: Features

Features: pieces of evidences describing some aspects of observed data x, usually with respect to the predicted label y

- computer vision
  - the shape, color, texture, size.....of an object
  - other objects nearby, relative postions
  - number of objects available
  - ...
- natural language processing
  - the target word itself, POS, prefix, suffix, capital or not, ...
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  - word embedding
  - tree embedding, graph embedding, ...
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  - embedding anything ...
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$\rightarrow$ **discriminative models**

Features in NLP

- A feature is **a function of** x **and** y**,** $f_i(x, y) \in \mathcal{R}$
- more often, it is a binary or indicator function:

Example

- when we do WSD for the target word *bank*,

$$f_i = \begin{cases} 1 & \text{if } w_{-1} = \text{transfer and } y = \textbf{FINANCIAL}, \\ 0 & \text{otherwise} \end{cases}$$

if the previous word is *transfer*, the current target should have the sense of **FINANCIAL**.

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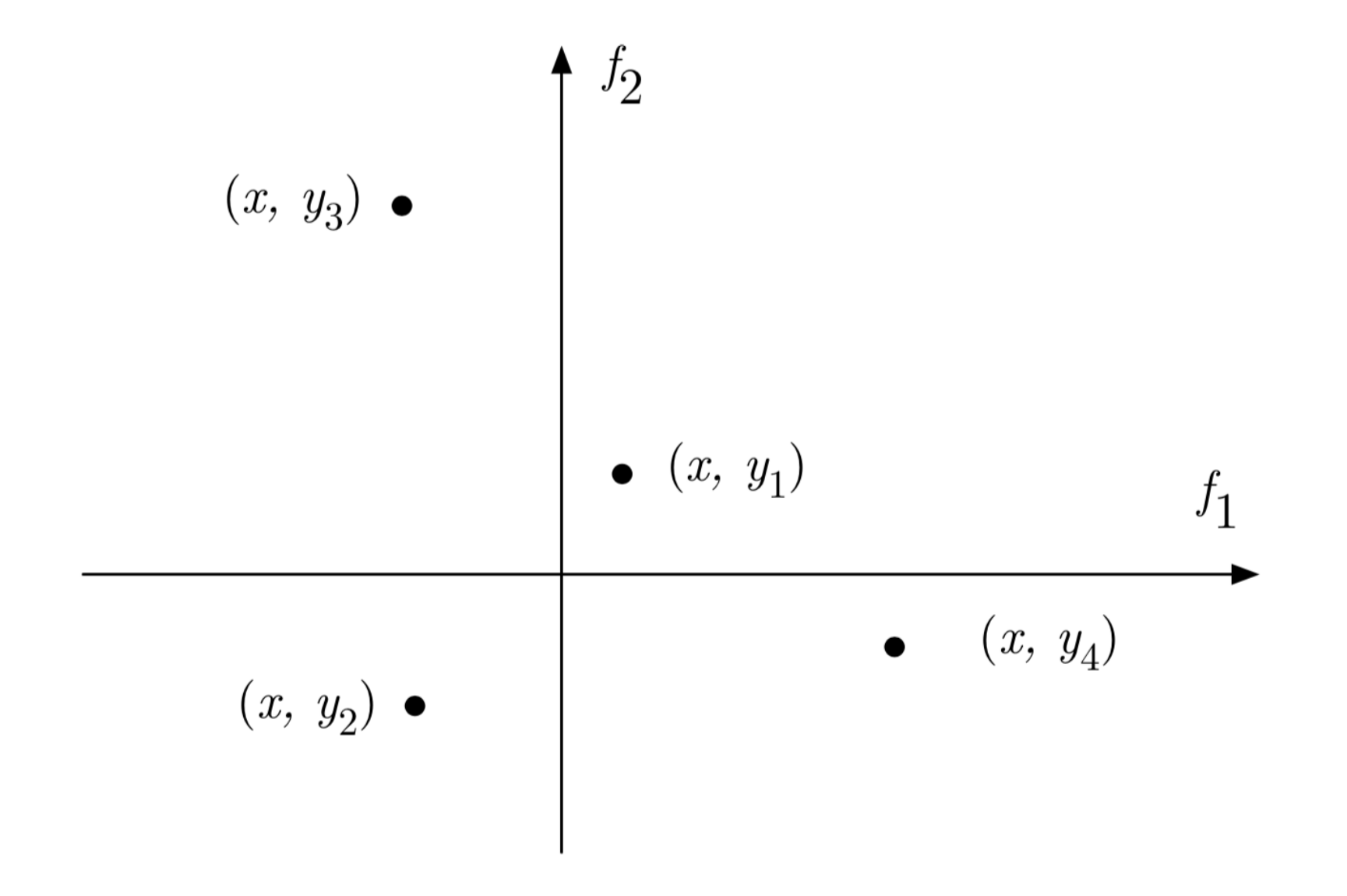
- if we have m aspects to describe an instance, i.e., m features, then:
  - **a feature vector** for each instance, (x, y)
  - $[f_1(x, y), f_2(x, y), f_3(x, y), ..., f_m(x, y)]$
  - $[1, 0, 0, ..., 1, 0]$

Features in NLP



[from Noah Smith]

Examples of Features in NLP

WSD for **bank**:

- if the previous word is transfer, it should be of **FINANCIAL**.
- if the next word is note, it should be of **FINANCIAL**.
- if the sentence is more than 10 words long, it should be of **FINANCIAL**.
- if **bank** is the first word of the sentence, it should be of **FINANCIAL**.
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Any differences compared to what we have done last week?

Naïve Bayes?

Examples of Features in NLP

Text Classification for news reports:

- if the document contains Kobe, its category should be of **Sports**
- if the title contains NBA, its category should be of **Sports**
- if football appear in the upper half, its category should be of **Sports**
- ...

An Example in WSD

I cash a check in that **bank** *and transfer 100 dollars to my mom.*

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Feature templates

- words before target words: $w_{-1} =$ *, $w_{-2} =$ *, $w_{-3} =$ *, ...
- words after target words: $w_1 =$ *, $w_2 =$ *, $w_3 =$ *, ...
- whether at the beginning of sent.: $@1 = \{$YES, NO$\}$,
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Apply those feature templates to the input sentence

- $w_{-1} =$ that, $w_{-2} =$ in, $w_{-3} =$ check, ...
- $w_1 =$ and, $w_2 =$ transfer, $w_3 =$ 100, ...
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when we consider a candidate label Fi for this case:

- $f_{w-1=that,Fi} = 1$, $f_{w-2=in,Fi} = 1$, $f_{w-3=check,Fi} = 1$, ...
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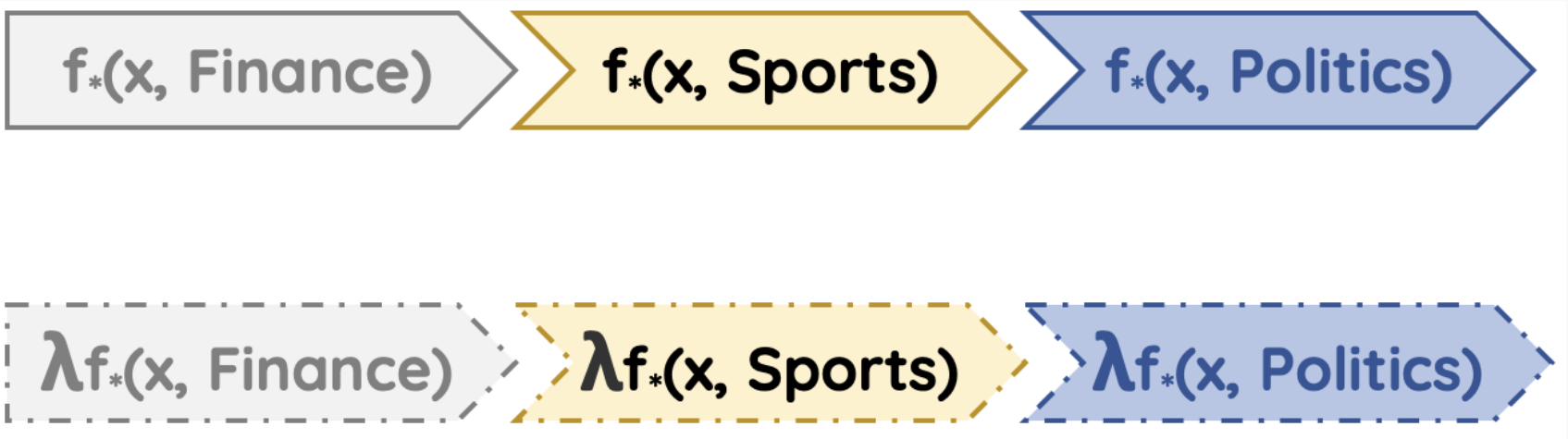
simply written as: $f(x, y)$

i.e., $f(x, Fi)$, $f(x, Sp)$, $f(x, Po)$, $f(x, En)$,...

Feature Vectors

The **long** feature vector will look like:



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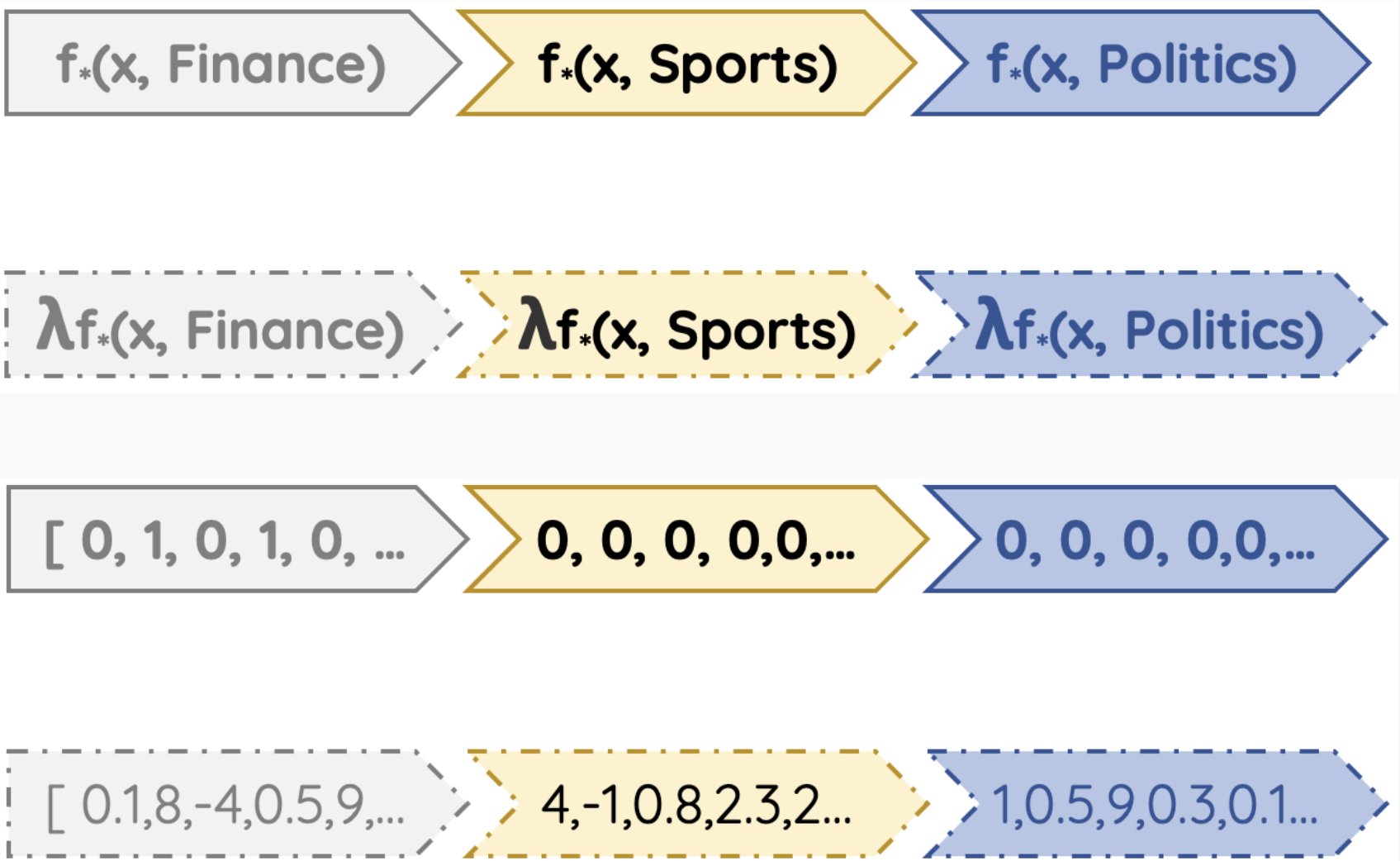

Feature Vector

Or, you can consider one feature vector and weight vector for each category:

Finance	Sports	Politics
0	0	0
1	1	1
0	0	0
1	1	1
0	0	0
0	0	0
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1	1	1
...	...	...

Feature Vector

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Finance	$\lambda_{Finance}$	Sports	λ_{Sports}	Politics	$\lambda_{Politics}$
0	0.1	0	4	0	1
1	8	1	-1	1	0.5
0	-4	0	0.8	0	9
1	0.5	1	2.3	1	0.3
0	9	0	2	0	0.1
0	-3	0	3	0	4
0	4	0	4	0	6
0	0.7	0	6	0	8.9
1	9	1	8.0	1	-2
...	...	...	...	...	...

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所有文章的个数

DF_t是出现该词语的文章个数

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Take an hour to research the TF-IDF weighting scheme

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$\Longrightarrow$:

- $[1, 1, 1, 1, 0, 0, ..., 1, 0, 0]$, also called One-Hot vector
- Or, using frequencies: $[1, 2, 1, 4, 0, 0, ..., 90, 0, 0]$
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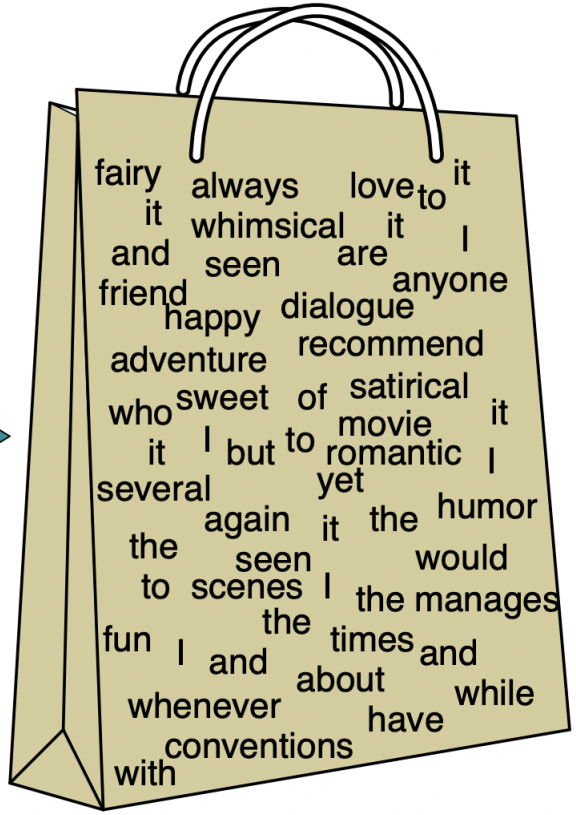
The Bag-of-words Representation

只统计数量、重要性，不考虑顺序

The so-called `Bag-of-words` format

I love this movie! It's sweet, but with satirical humor. The dialogue is great and the adventure scenes are fun... It manages to be whimsical and romantic while laughing at the conventions of the fairy tale genre. I would recommend it to just about anyone. I've seen it several times, and I'm always happy to see it again whenever I have a friend who hasn't seen it yet!



it	6
I	5
the	4
to	3
and	3
seen	2
yet	1
would	1
whimsical	1
times	1
sweet	1
satirical	1
adventure	1
genre	1
fairy	1
humor	1
have	1
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...	...

[Jurafsky and Martin, SLP3]

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What are those with very small values or even zeros?

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A Simple Features based Discriminative Model

Imagine a linear classifiers with the form like, $\lambda_{f(x,y)} f(x,y)$, where λs are weights,

- build a linear function to map $f(x,y)$ to label y
- possibly need a weight $\lambda_{f_i(x,y)}$ for each feature $f_i(x,y)$
- then, for each possible label y of instance x, we can compute a score:

$$score(x,y) = \sum_i \lambda_{f_i(x,y)} f_i(x,y)$$

- the classifier should choose y^*:

$$y^* = \arg\max_y \sum_i \lambda_{f_i(x,y)} f_i(x,y)$$

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That is, for each y, compute its score, and select the y^* that gives the largest score.

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Note that it may not be a probabilistic model.

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$$y^* = \arg\max_y \sum_i \lambda_{f_i(x,y)} f_i(x,y)$$

How to figure out those λs?

Our Expectations

We want:

- nicely fit to our linear story
- (at least, looks like) simple
- (may be) easy to drive
- (hopefully) probabilistic
- ...

Features based Linear Models: Algorithms

The key is to choose/learn proper weights λs for different features

- the Perceptron algorithm
- Margin-based models (the Support Vector Machines, SVM)
- Exponential Models:
  - log-linear models, maximum entropy models, logistic models, ...

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$$p(y|x) = \frac{\exp score(x, y)}{\sum_{y'} \exp score(x, y')} = \frac{\exp \sum_i \lambda_{f_i(x,y)} f_i(x, y)}{\sum_{y'} \exp \sum_i \lambda_{f_i(x,y')} f_i(x, y')}$$

  - numerator: positive score for label y
  - denominator: normalization over all labels

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  - numerator: positive score for label y
  - denominator: normalization over all labels
  - everything is positive; looks like a probabilistic interpretation!
  - a very powerful tool!

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Log-Linear Models

For a data sample (x, y);

- We care:

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目标：令\hat{y}
在x条件下概率最高 要变的参数是\lambda

- linear: $\boldsymbol{\lambda} \cdot \boldsymbol{f}(x,y)$
- Normalization: $\log \sum_{y'} \exp(\boldsymbol{\lambda} \cdot \boldsymbol{f}(x,y'))$

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Maximum-Likelihood Estimations

How likely do we observe the data given the current parameters?

- Given the training data $\{(x_1, y_1), (x_2, y_2), ..., (x_k, y_k)\}$: the Likelihood of $\boldsymbol{\lambda}$ is:

$$L(\boldsymbol{\lambda}) = \prod_k p(y_k|x_k; \boldsymbol{\lambda}) = \prod_k \frac{\exp(\boldsymbol{\lambda} \cdot \boldsymbol{f}(x_k, y_k))}{\sum_{y'} \exp(\boldsymbol{\lambda} \cdot \boldsymbol{f}(x_k, y'))}$$

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  - The first component: **Empirical Counts**
  - The second component: **Expected Counts**

Model Training

In feature-based Log-Linear models,

- "We hope **Empirical Counts** equals **Expected Counts**"

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  - volume
  - coverage
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  - ...

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Gradient Ascend Methods

- Need to maximize $LL(\boldsymbol{\lambda})$ where

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- imagine: treat your training set as a very large batch
- Slow!
  - Optimizations available : Conjugate Gradient Methods
  - or, Stochastic Gradient Methods

Stochastic Gradient Ascend

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- Initialize all λs, number of epochs T, learning rate α
- For $t \in \{1, ..., T\}$:
  - choose a random permutation π of $\{1, 2, ..., k, ...\}$ (the whole dataset)
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imagine: treat each instance as a small batch

More to Mention

Just anther way of talking this:

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- then, use Stochastic Gradient **Descend** instead of *Ascend* in the last slide
- all seem exciting terms :-)

Anything More?

Our main goal is to get the $\boldsymbol{\lambda}s$:

- How do we get the probablistic thing?
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- Do you believe we can do anything with the features?

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It looks we have everything ready to build a model!

- Say, we have a feature f_1, defined as:

$$f_1(x, y) = \begin{cases} 1 & \text{if } x \text{ contains } \textbf{NFL} \text{ and } y = \textbf{Sports}, \\ 0 & \text{otherwise} \end{cases}$$

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Regularization

What can we do to prevent such cases?

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$\rightarrow$ Regularize the learning process

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→ Regularize the learning process

- Modified the loss function

$$LL(\boldsymbol{\lambda}) = \sum_k \boldsymbol{\lambda} \cdot \boldsymbol{f}(x_k, y_k) - \sum_k \log \sum_{y'} \exp(\boldsymbol{\lambda} \cdot \boldsymbol{f}(x_k, y')) - \frac{\alpha}{2} ||\boldsymbol{\lambda}||^2$$

$$= \sum_k \boldsymbol{\lambda} \cdot \boldsymbol{f}(x_k, y_k) - \sum_k \log \sum_{y'} \exp(\boldsymbol{\lambda} \cdot \boldsymbol{f}(x_k, y')) - \frac{\alpha}{2} \sum_i \lambda_i^2$$

- When calculating the gradients

$$\frac{\partial LL(\boldsymbol{\lambda})}{\partial \lambda_{f_i(x,y)}} = \sum_k f_i(x_k, y_k) - \sum_k \sum_{y'} f_i(x_k, y') p(y' | x_k; \boldsymbol{\lambda}) - \alpha \lambda_{f_i(x,y)}$$

- Adds a penalty for large weights

Regularization

Prevent every parameter from becoming too large in magnitude.

$$\arg\min_{\boldsymbol{\lambda}} loss(\boldsymbol{\lambda}) + \alpha||\boldsymbol{\lambda}||_p$$

where $\alpha > 0$, p could choose from 1, 2 or others.

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Gives us:

- L-2 Regularization
- L-1 Regularization
  - Sparsity! (parameters)
  - but .....

Evaluation

How well does your model perform?

- Evaluation Metrics
- Evaluation Protocols
  - Training dataset
  - Test dataset
  - Development set

Reasons for Log-linear Models

You will find it useful in the (near) future...

- You love feature engineering;
- You hate feature engineering
- Building blocks for future research

In Short

- A strong classifier you would be more familiar in the future
  - connection to many dominant classifiers
- Take care of features for your tasks
  - feature selection
  - biased feature selection
  - purposely feature selection
- Take care of the learning process and also evaluation protocal.
  - regularization
  - randomness

Reading

Book Chapter 2 and 4, Dan's book

Notes M. Collins, Notes on Log-Linear Models (http://www.cs.columbia.edu/ mcollins/loglinear.pdf)

Notes Lecture Note, Linear Regression and Gradient Ascent, CS109@Stanford, (https://web.stanford.edu/class/archive/cs/cs109/cs109.1208/lectures/2

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Further Reading

- David M. Blei, Andrew Y. Ng and Michael I. Jordan. Latent Dirichlet Allocation, Journal of Machine Learning Research 3 (2003) 993-1022
- Zhang and Oles (2010), Text Categorization Based on Regularized Linear Classification Methods (http://www.stat.yale.edu/lc436/papers/temp/Zhang Oles 2001.pdf)
- Lecture Notes 1, Discriminative Algorithms, CS229-Machine Learning@Stanford, Andrew Ng (http://cs229.stanford.edu/materials.html)
- Adam Berger, Stephen Della Pietra, and Vincent Della Pietra, A maximum entropy approach to natural language processing. Computational Linguistics, 22(1):39-71, 1996.
- Galen Andrew and Jianfeng Gao, Scalable training of L1-regularized log-linear models. In Proc. of ICML, 2007.